

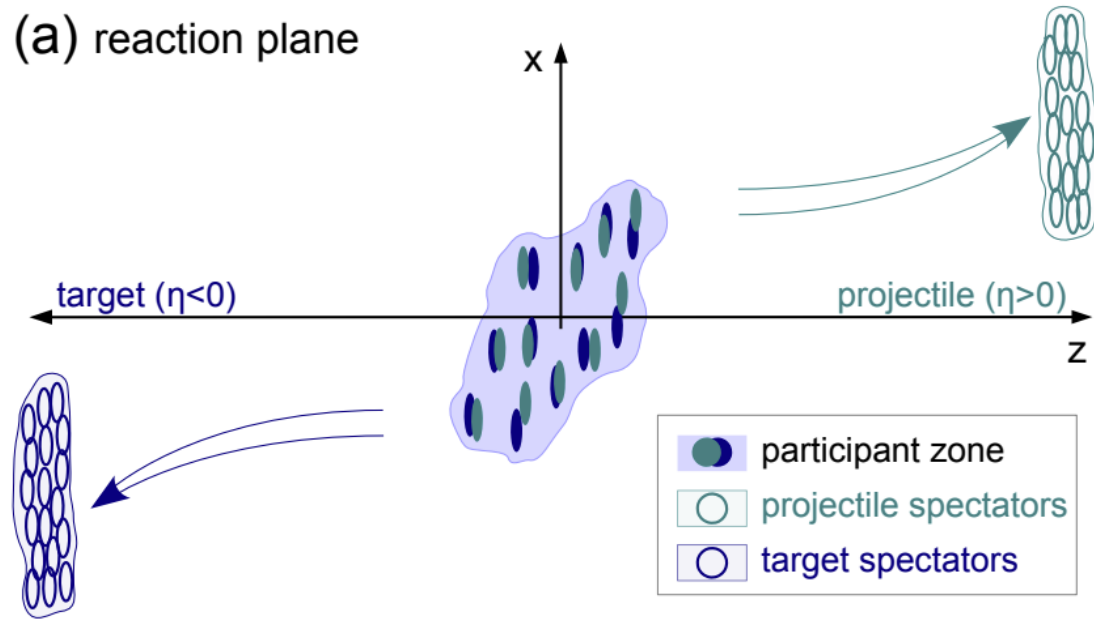
Directed flow and global polarization in Au+Au collisions

<https://arxiv.org/pdf/2301.02960>

Ze-Fang Jiang, Xiang-Yu Wu, Shanshan Cao, and Ben-Wei Zhang

Motivation

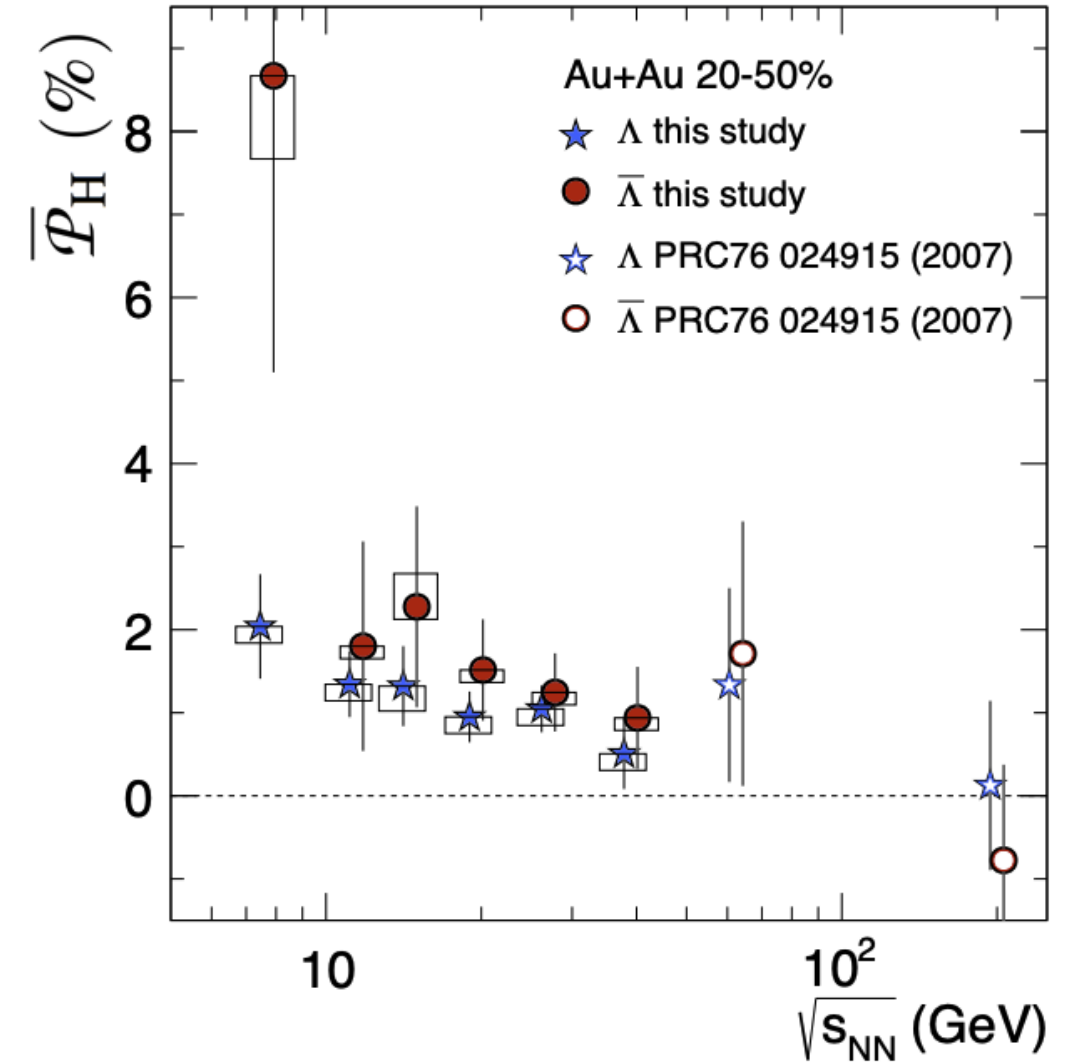
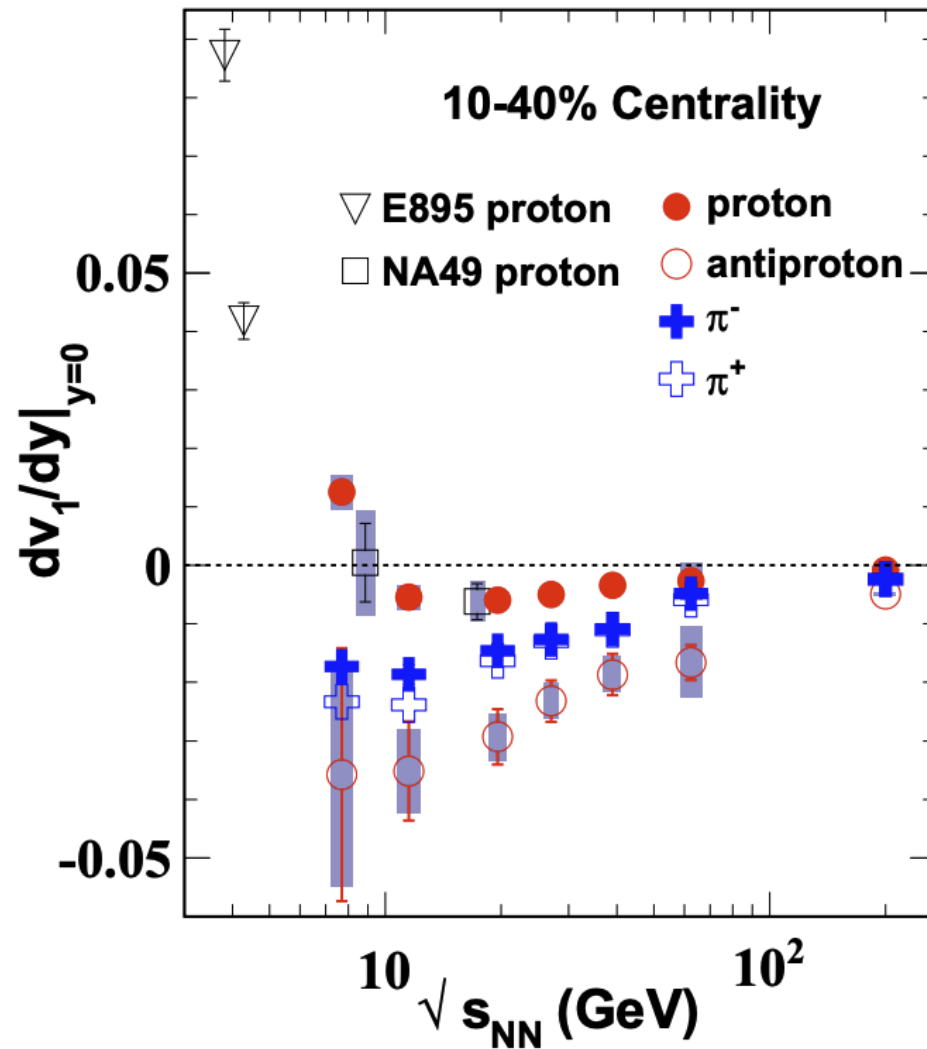
(a) reaction plane



Schematic of collision geometries for directed flow (left) and global Λ -hyperon polarization (right)

- i) In a noncentral heavy-ion collision, the initial geometry is not symmetric in the reaction plane $\rightarrow$ directed flow.
- ii) The global angular momentum $\mathbf{J}$ is perpendicular to the reaction plane; the spin of the Λ hyperon aligns with $\mathbf{J} \rightarrow$ polarization.

Beam-energy dependence of dv_1/dy at $y = 0$ and $\bar{P}_y$



What's done ?

- Study Au+Au collisions at $\sqrt{s_{NN}} = 7.7\text{--}62.4$ GeV
- Model a tilted QGP fireball and longitudinal flow
- Describe particle spectra and directed flow
- Test the same initial condition with global Λ and Λ^- polarization

Model framework

Generating tilted initial geometry from modified Glauber model

$$T_1(\mathbf{x}_T) = T_+(\mathbf{x}_T) \left\{ 1 - \left[1 - \frac{\sigma_{NN} T_-(\mathbf{x}_T)}{A} \right]^A \right\}$$

$$T_2(\mathbf{x}_T) = T_-(\mathbf{x}_T) \left\{ 1 - \left[1 - \frac{\sigma_{NN} T_+(\mathbf{x}_T)}{A} \right]^A \right\}$$

$$W_N(x, y, \eta_s) = T_1(x, y) + T_2(x, y) + H_t [T_1(x, y) - T_2(x, y)] \tan \left(\frac{\eta_s}{\eta_t} \right)$$

- H_t : tilt strength
- $\tan(\eta_s/\eta_t)$: rapidity dependence
- $\eta_t = 8.0$: fixed parameter

$$T(x, y) = \int_{-\infty}^{\infty} dz \frac{n_0}{1 + \exp \left(\frac{r - R_0}{d_0} \right)}, \quad r = \sqrt{x^2 + y^2 + z^2}$$

Model framework

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$$W_N(x, y, \eta_s) = T_1(x, y) + T_2(x, y) + H_t [T_1(x, y) - T_2(x, y)] \tan \left(\frac{\eta_s}{\eta_t} \right)$$

$$W(x, y, \eta_s) = \frac{(1 - \alpha)W_N(x, y, \eta_s) + \alpha n_{\text{BC}}(x, y)}{[(1 - \alpha)W_N(0, 0, 0) + \alpha n_{\text{BC}}(0, 0)]_{b=0}}$$

- n_{BC} : number of binary collisions
- $\alpha = 0.05$: determined from the centrality dependence of the hadron yield

Initial Energy and Baryon Densities

$$\varepsilon(x, y, \eta_s) = K W(x, y, \eta_s) H(\eta_s)$$

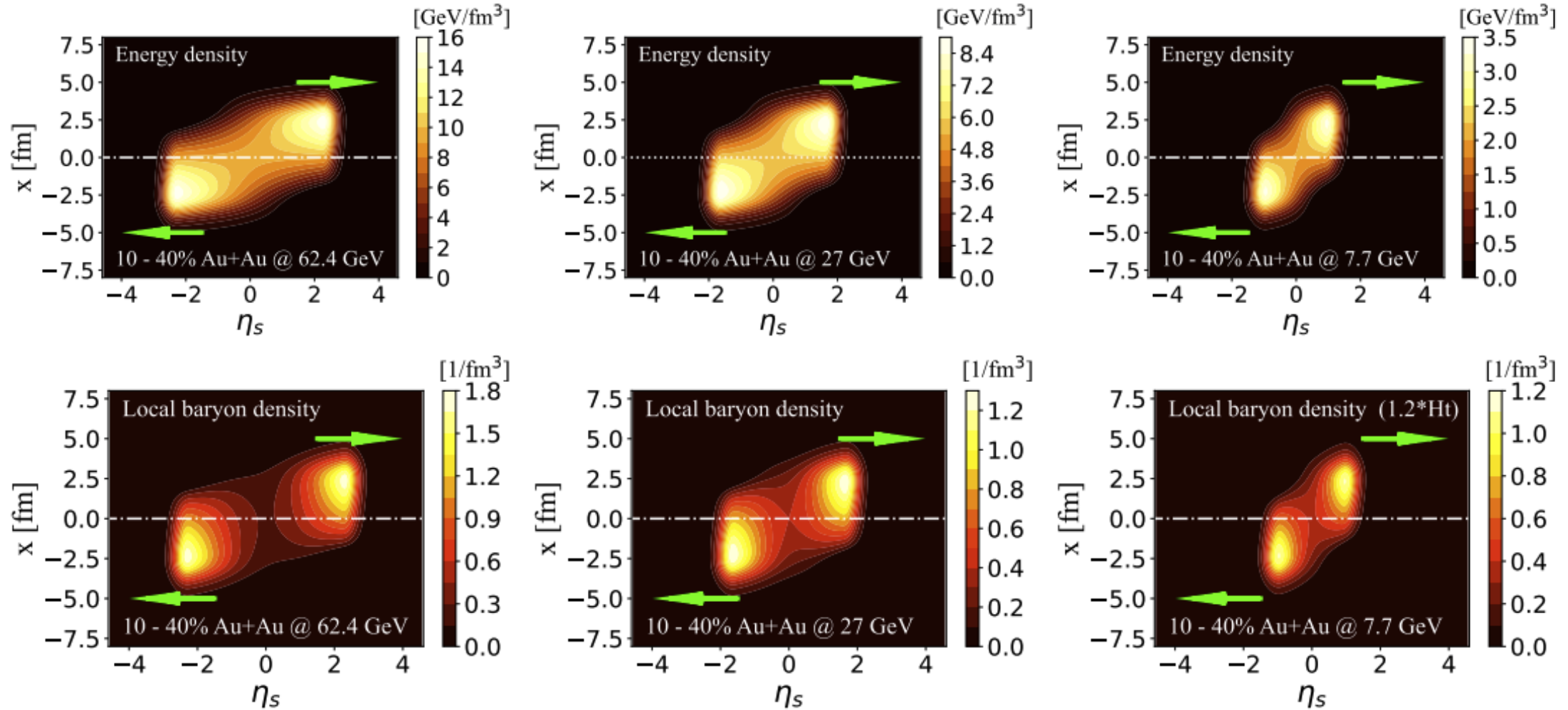
$$n(x, y, \eta_s) = \frac{1}{N} W(x, y, \eta_s) H(\eta_s) H_B(\eta_s)$$

$$H(\eta_s) = \exp \left[-\frac{(|\eta_s| - \eta_w)^2}{2\sigma_\eta^2} \theta(|\eta_s| - \eta_w) \right]$$

$$H_B(\eta_s) = \exp \left[-\frac{(\eta_s - \eta_n)^2}{2\sigma_n^2} \right] + \exp \left[-\frac{(\eta_s + \eta_n)^2}{2\sigma_n^2} \right]$$

- $K \leftarrow dN_{\text{ch}}/d\eta$ or dN_{ch}/dy
- $N \leftarrow N_{\text{part}}$
- $\eta_n, \sigma_n \leftarrow p_T$ spectra of $p, \bar{p}$

Initial ε and n distributions in the η_s - x plane



Parametrizing Longitudinal Flow

$$T^{\tau\tau} = \varepsilon(x, y, \eta_s) \cosh(y_L)$$

$$T^{\tau\eta_s} = \frac{1}{\tau_0} \varepsilon(x, y, \eta_s) \sinh(y_L)$$

$$y_L \equiv f_v y_{\text{CM}}$$

$$y_{\text{CM}} = \text{arctanh} \left[\frac{T_1 - T_2}{T_1 + T_2} \tanh(y_{\text{beam}}) \right]$$

- $f_v \in [0, 1]$: longitudinal momentum fraction
- $f_v = 0 \rightarrow y_L = 0$: Bjorken flow
- $y_{\text{beam}} = \text{arccosh}[\sqrt{s_{NN}}/(2m_N)]$
- Transverse components: $T^{\tau x} = T^{\tau y} = 0$

Hydrodynamic Evolution and Particlization

(3+1)D viscous hydrodynamics — CLVisc

$$\nabla_\mu T^{\mu\nu} = 0, \quad \nabla_\mu J^\mu = 0$$

$$T^{\mu\nu} = \varepsilon U^\mu U^\nu - P \Delta^{\mu\nu} + \pi^{\mu\nu}$$

$$J^\mu = n U^\mu + V^\mu$$

$$\Delta^{\mu\nu} = g^{\mu\nu} - U^\mu U^\nu, \quad g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

- $\pi^{\mu\nu}$: shear-viscous stress
- V^μ : baryon diffusion current
- Israel–Stewart second-order evolution

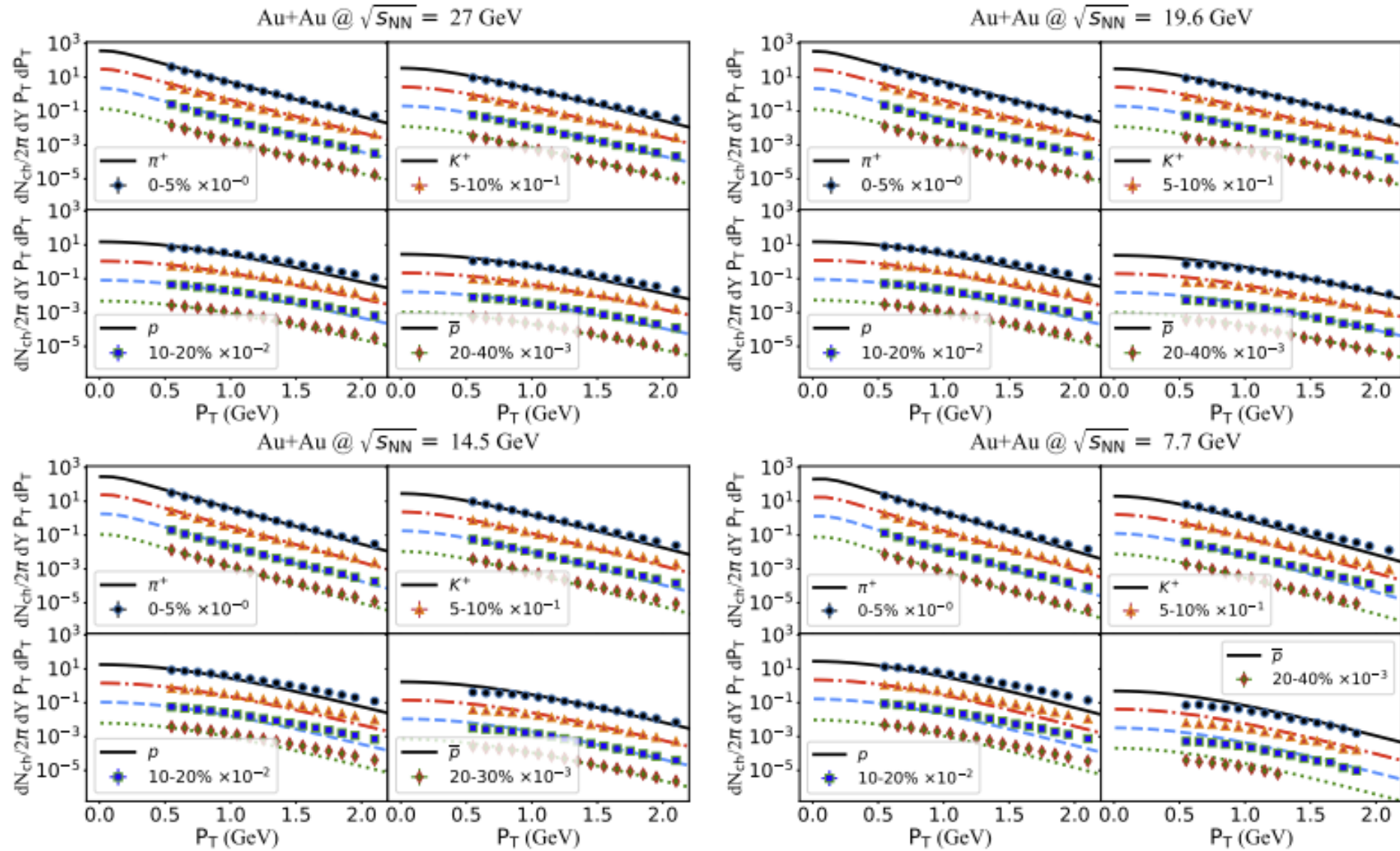
Cooper–Frye particlization

$$\frac{dN}{p_T dp_T d\phi dy} = \frac{g_i}{(2\pi)^3} \int_\Sigma p^\mu d\Sigma_\mu f_{\text{eq}} (1 + \delta f_\pi + \delta f_V)$$

$$\varepsilon_{\text{frz}} = 0.4 \text{ GeV/fm}^3$$

RESULTS

Identified particle spectra



Parameters of the initial state model

$\sqrt{s_{\text{NN}}}$ [GeV]	K	τ_0 [fm]	σ_η [fm]	η_w	σ_n	η_n	f_v	H_t
62.4	11.8	1.0	0.3	2.25	1.34	2.7	0.18	10.0
39	8.25	1.3	0.3	1.9	1.13	2.1	0.22	13.0
27	7.40	1.4	0.3	1.6	1.06	1.8	0.23	13.5
19.6	5.60	1.8	0.3	1.3	0.85	1.5	0.24	15.5
14.5	3.90	2.2	0.3	1.15	0.81	1.4	0.24	18.0
11.5	3.05	2.4	0.3	1.16	0.79	1.22	0.25	22.0
7.7	2.50	2.6	0.3	0.9	0.70	1.05	0.26	28.0

First six parameters

→ charged-particle
yield +
proton/antiproton p_t
spectra

• **f_v and H_t** →
insensitive to ϕ -
integrated spectra;
constrained by v_1

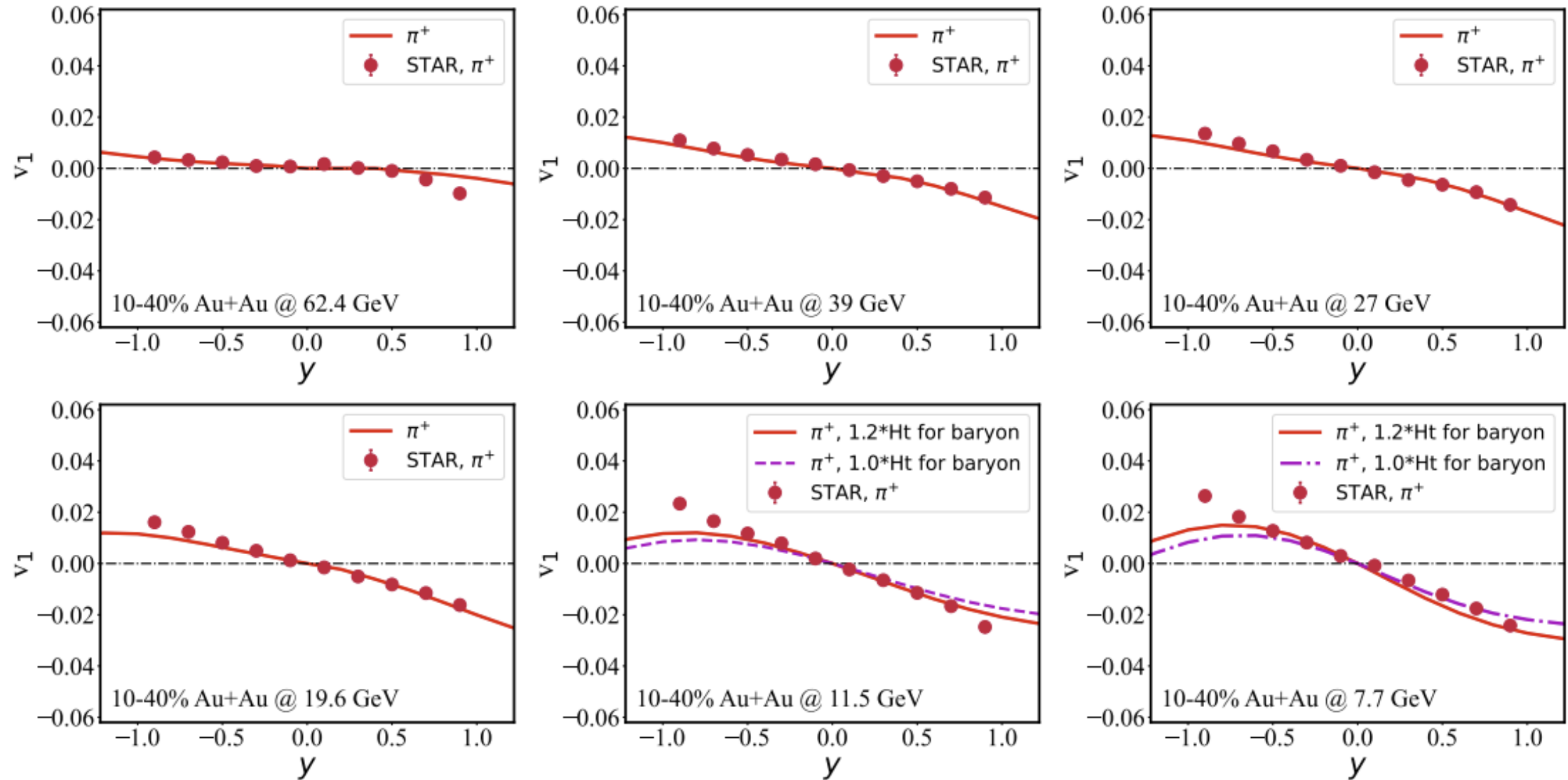
TABLE II: Parameters for the 3-dimensional optical Glauber model

Directed Flow Coefficient $V_1(y)$

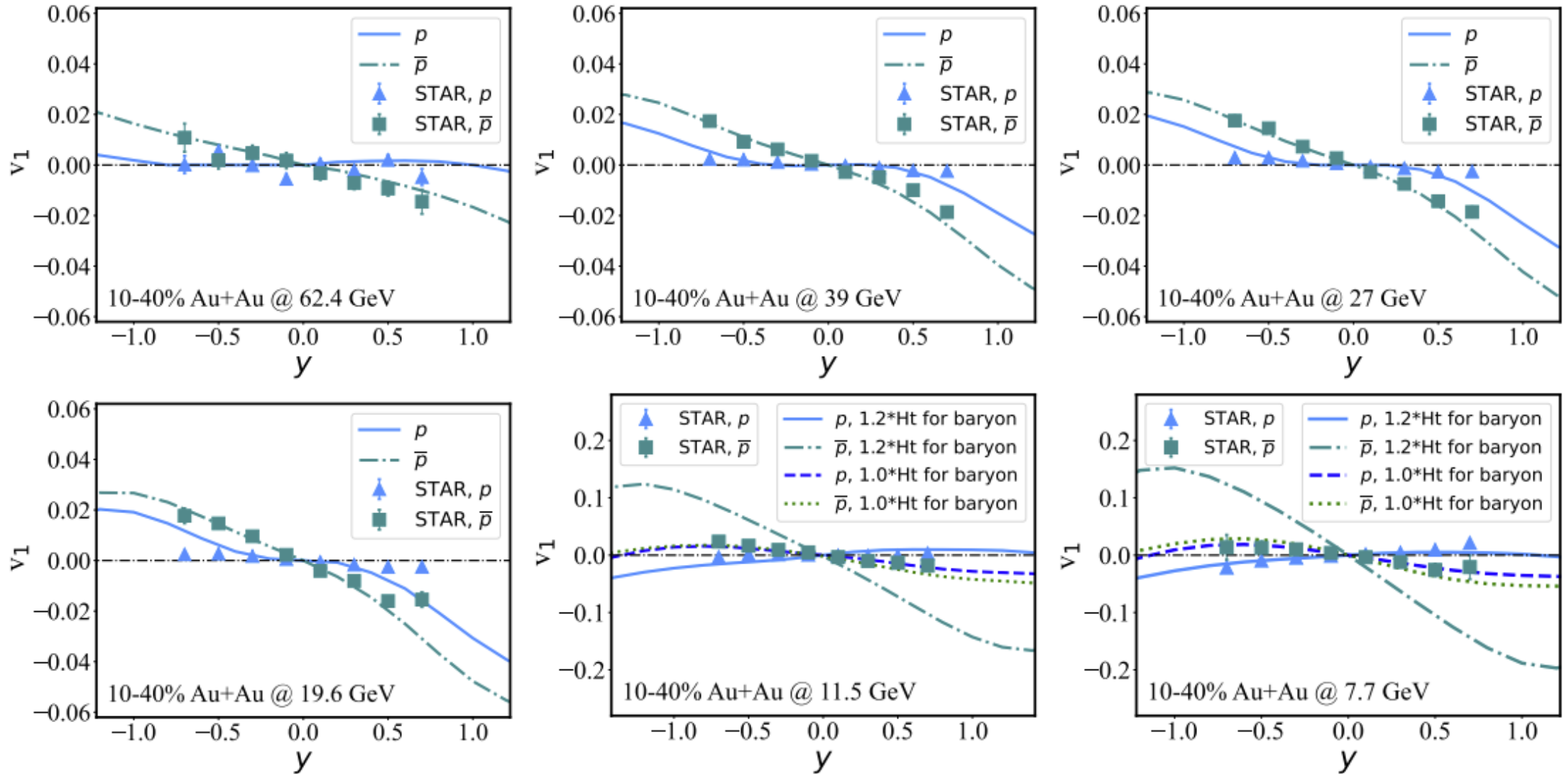
$$v_1(y) = \langle \cos(\phi - \Psi_1) \rangle = \frac{\int \cos(\phi - \Psi_1) \frac{dN}{dy d\phi} d\phi}{\int \frac{dN}{dy d\phi} d\phi}$$

- Ψ_1 : first-order event-plane angle
- Smooth initial condition $\rightarrow$ event plane = spectator plane
- π^+ , p , $\bar{p}$ in 10–40% Au+Au collisions
- Soft hadrons: $0 < p_T < 3.0$ GeV
- H_t , f_v : constrained by $v_1(y)$
- 11.5 and 7.7 GeV: $1.2H_t$ for the baryon-density profile

Rapidity dependence of the directed flow coefficient of π^+



Rapidity dependence of the directed flow coefficient of protons and antiprotons



Slope of rapidity dependent directed flow across collision energies

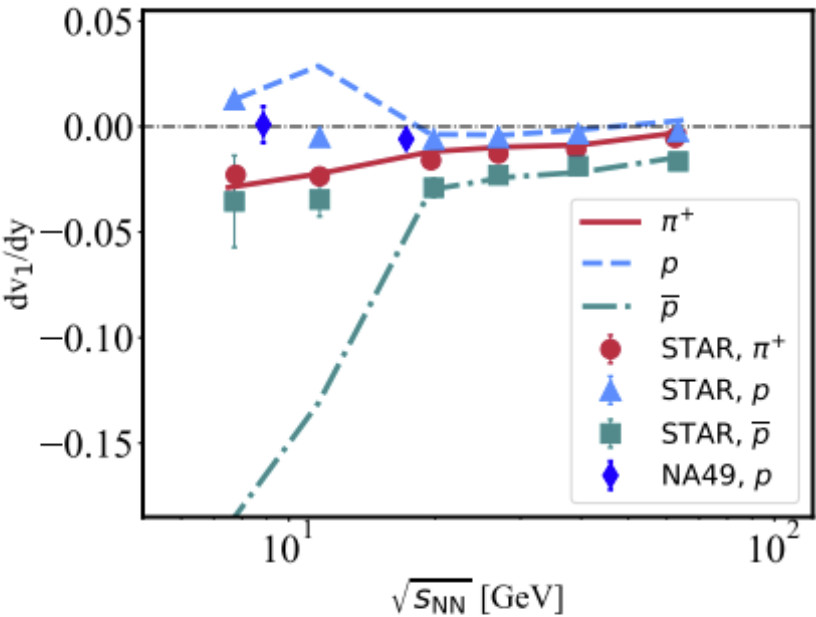
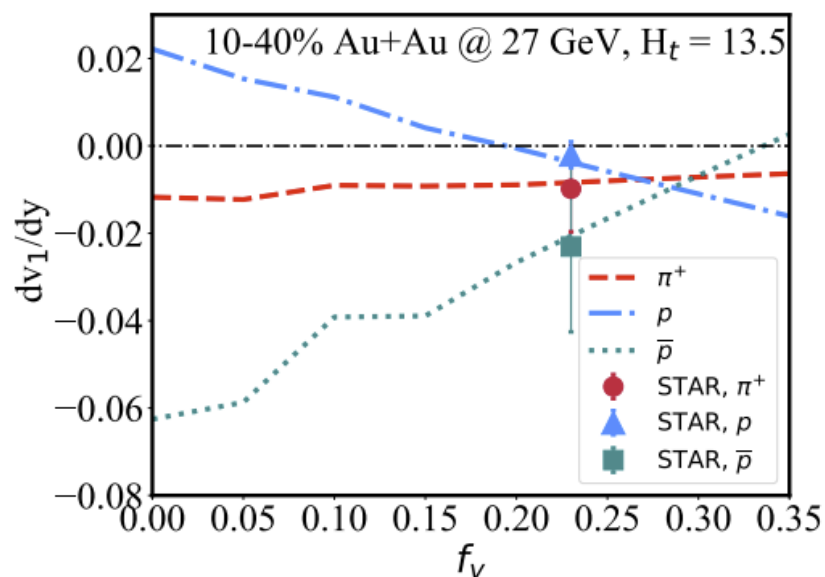
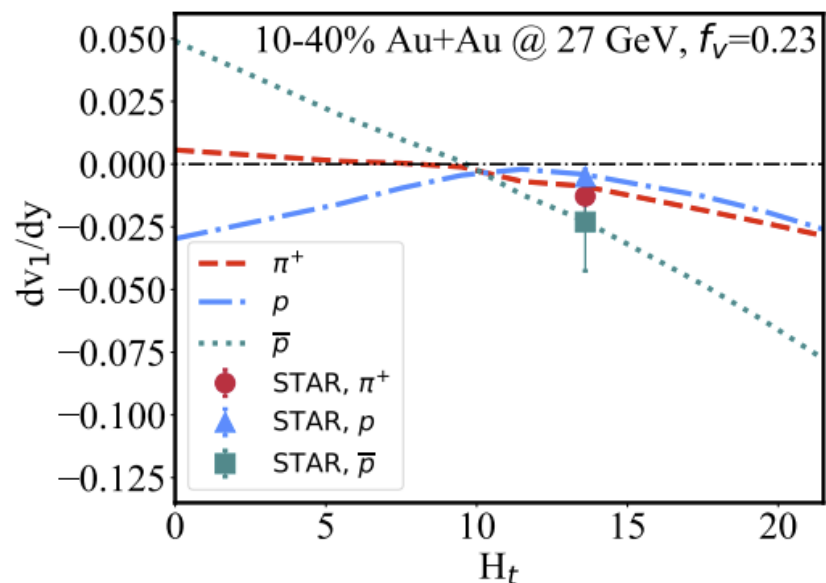


FIG. 5: (Color online) The slope of the rapidity dependence of the directed flow coefficients of pions, protons and antiprotons as functions of the colliding energy in 10-40% Au+Au collisions. Results from our model calculation are compared to the STAR [19] and NA49 [51] data. Here, $1.2H_t$ is applied to the baryon number density distribution at $\sqrt{s_{NN}} = 11.5$ and 7.7 GeV.

f_v	H_t
0.18	10.0
0.22	13.0
0.23	13.5
0.24	15.5
0.24	18.0
0.25	22.0
0.26	28.0

- Higher $\sqrt{s_{nn}} \rightarrow$ smaller proton–antiproton v_1 splitting
- Weaker baryon-density tilt + smaller longitudinal momentum fraction
- Lower $\sqrt{s_{nn}} \rightarrow$ larger H_t and $f_v \rightarrow$ larger v_1 splitting

Determination of H_t and f_v from Directed Flow



- With finite initial longitudinal flow and **small H_t** , pion and antiproton v_1 is positive at forward rapidity and negative at backward rapidity; the **sign flips when H_t is large**
- Pions and antiprotons mainly follow the energy-density distribution, while **protons carry baryons deposited by the beam nuclei** and are strongly **affected by the initial baryon-density distribution**
- The pion and antiproton **dv_1/dy decreases as H_t increases**; the opposite behavior is observed as f_v increases
- At $\sqrt{s_{nn}} = 27$ GeV, $H_t = 13.5$ and $f_v = 0.23$ provide the best simultaneous description of pion, proton, and antiproton v_1

Global polarization

$$S^\mu(\mathbf{p}) = \frac{\int d\Sigma \cdot p \mathcal{J}_5^\mu(p, X)}{2m \int d\Sigma \cdot \mathcal{N}(p, X)}$$

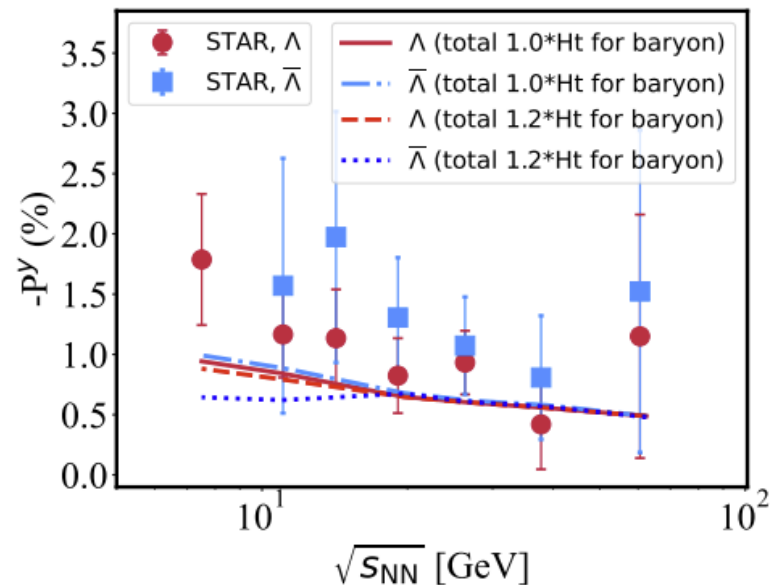
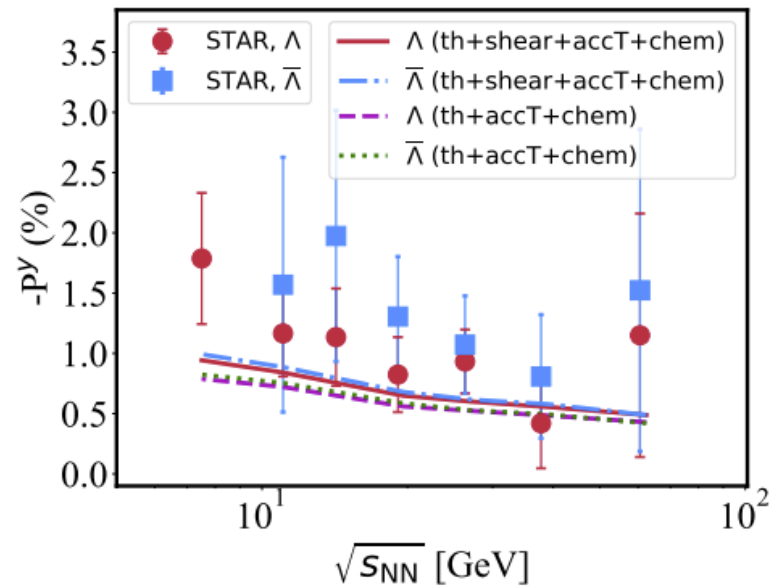
$$S^\mu = S_{\text{thermal}}^\mu + S_{\text{shear}}^\mu + S_{\text{accT}}^\mu + S_{\text{chemical}}^\mu + S_{\text{EB}}^\mu$$

$$P^\mu(\mathbf{p}) = \frac{1}{s} S^\mu(\mathbf{p}), \quad s = \frac{1}{2}$$

$$\mathbf{P}^*(\mathbf{p}) = \mathbf{P}(\mathbf{p}) - \frac{\mathbf{P}(\mathbf{p}) \cdot \mathbf{p}}{p^0(p^0 + m)} \mathbf{p}$$

$$\langle \mathbf{P}(\phi_p) \rangle = \frac{\int_{y_{\min}}^{y_{\max}} dy \int_{p_{T \min}}^{p_{T \max}} p_T dp_T [\Phi(\mathbf{p}) \mathbf{P}^*(\mathbf{p})]}{\int_{y_{\min}}^{y_{\max}} dy \int_{p_{T \min}}^{p_{T \max}} p_T dp_T \Phi(\mathbf{p})}$$

Global polarization of Λ and $\bar{\Lambda}$



- Thermal vorticity + shear + fluid acceleration/temperature gradient + chemical-potential gradient
- Electromagnetic-field contribution neglected
- H_t and f_v from $v_1 \rightarrow$ reasonable description of Λ and $\bar{\Lambda}$ polarization
- Total polarization $\rightarrow$ mainly thermal-vorticity contribution
- Counterclockwise tilt $\rightarrow$ negative shear tensor $\rightarrow$ larger magnitude of $-P_Y$
- Stronger baryon-density tilt at 11.5 and 7.7 GeV $\rightarrow$ positive chemical contribution to $P_Y \rightarrow$ smaller polarization

Conclusion

- Modified Glauber model: tilted energy-density and net-baryon-density profiles + initial longitudinal flow
- Combined π , p , and $\bar{p}$ v_1 : constrain initial geometry and longitudinal flow through H_t and f_v , which increase toward lower collision energies
- $H_t \uparrow$: $dv_1/dy \downarrow$; $f_v \uparrow$: $dv_1/dy \uparrow$; proton v_1 strongly affected by the initial baryon-density distribution
- Λ and Λ^- global polarization: reasonably described within the same framework

Other references

- Image references in the introduction
- Directed flow - <https://arxiv.org/pdf/1306.4145>
- Polarization - <https://arxiv.org/pdf/1701.06657>

Additional slides

$$\mathcal{S}^\mu(\mathbf{p}) = \mathcal{S}_{\text{thermal}}^\mu(\mathbf{p}) + \mathcal{S}_{\text{shear}}^\mu(\mathbf{p}) + \mathcal{S}_{\text{accT}}^\mu(\mathbf{p}) \\ + \mathcal{S}_{\text{chemical}}^\mu(\mathbf{p}) + \mathcal{S}_{\text{EB}}^\mu(\mathbf{p}),$$

where

$$\mathcal{S}_{\text{thermal}}^\mu(\mathbf{p}) = \int d\Sigma^\sigma F_\sigma \epsilon^{\mu\nu\alpha\beta} p_\nu \partial_\alpha \frac{u_\beta}{T},$$

$$\mathcal{S}_{\text{shear}}^\mu(\mathbf{p}) = \int d\Sigma^\sigma F_\sigma \frac{\epsilon^{\mu\nu\alpha\beta} p_\nu u_\beta}{(u \cdot p)T} \\ \times p^\rho (\partial_\rho u_\alpha + \partial_\alpha u_\rho - u_\rho D u_\alpha),$$

$$\mathcal{S}_{\text{accT}}^\mu(\mathbf{p}) = - \int d\Sigma^\sigma F_\sigma \frac{\epsilon^{\mu\nu\alpha\beta} p_\nu u_\alpha}{T} \left(D u_\beta - \frac{\partial_\beta T}{T} \right)$$

$$\mathcal{S}_{\text{chemical}}^\mu(\mathbf{p}) = 2 \int d\Sigma^\sigma F_\sigma \frac{1}{(u \cdot p)} \epsilon^{\mu\nu\alpha\beta} p_\alpha u_\beta \partial_\nu \frac{\mu}{T},$$

$$\mathcal{S}_{\text{EB}}^\mu(\mathbf{p}) = 2 \int d\Sigma^\sigma F_\sigma \left[\frac{\epsilon^{\mu\nu\alpha\beta} p_\alpha u_\beta E_\nu}{(u \cdot p)T} + \frac{B^\mu}{T} \right],$$

(3)

with

$$F^\mu = \frac{\hbar}{8m_\Lambda \Phi(\mathbf{p})} p^\mu f_{\text{eq}} (1 - f_{\text{eq}}),$$

$$\Phi(\mathbf{p}) = \int d\Sigma^\mu p_\mu f_{\text{eq}}.$$